

Deep Hedging – Learning to Trade

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What we do before?

Equation: Optimal hedging

$$\pi(X) := \inf_{\underline{\delta} \in \mathcal{H}} \rho(X + (\underline{\delta} \cdot \underline{S})_T - C_T(\delta))$$

- ρ is risk measure.
- X is our asset.
- δ is weight.
- $(\underline{\delta} \cdot \underline{S})_T$ trade gain or loss.
- $C_T(\delta)$ is transaction cost.

What is the advantage of deep hedging?

- Model-free approach
- Considers transaction costs
- Applicable to any type of option
- Considers market impact

What we do this semester?

- Consider different α impact deep hedging.
- Consider different epoch impact deep hedging.
- Consider different transaction costs impact weight and PnL of deep hedging.
- Use real historical data into my hedging strategy and discuss whether this hedging strategy's effectiveness is significantly improved.

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BS-model Setting

- stock paths are under BS-model, The initial price is 100, strike price is 100, $\mu = 0$, $\sigma = 0.2$, and generate 5000 paths for our experiment.
- CVaR's $\alpha = 0.95$ and training iteration 1000
- Entropic's $\lambda = 0.95$ and training iteration 10000
- We test 3 different transaction cost ratios 0, 0.001425, and 0.01.

History data model Setting

- Take historical price data for Apple Inc. (AAPL) from January 1, 2012, to January 1, 2024.
- The data was divided into sequential groups (e.g. days 1-30, 2-31, etc.).
- The initial 2500 groups were allocated for training purposes, with the remainder reserved for testing.
- Each data set was normalized to commence at a price of 100.
- Use the σ calculated from the previous 100 days of historical data for each data as the market σ for the next period.
- We test 3 different transaction cost ratios 0, 0.001425, and 0.01.

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Deep hedging(CVaR) with different α

Our utility functions are

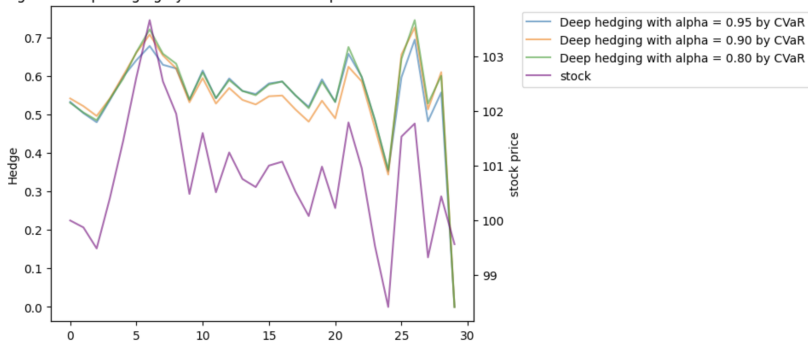
$$CVaR_{\alpha}(X) := \mathbb{E}[X | X \leq VaR_{\alpha}(X)]$$

where $VaR_{\alpha}(X) := P[X \leq \cdot]^{-1}(1 - \alpha)$

We compare with variable $\alpha = 0.95, 0.90, 0.80$

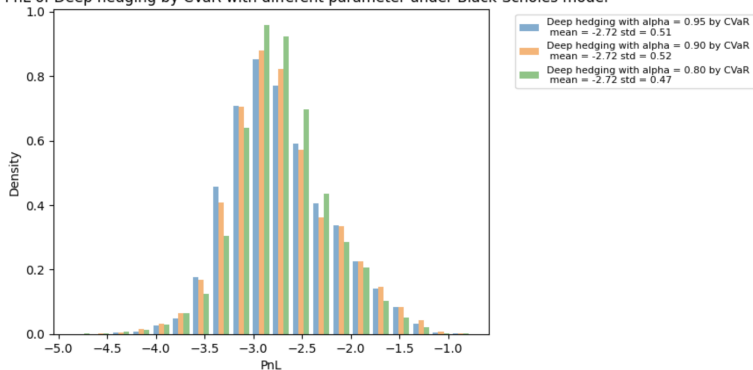
Deep hedging(CVaR) with different α

Compare weight of Deep hedging by CVaR with different parameter under Black-Scholes model



Deep hedging(CVaR) with different α

Compare PnL of Deep hedging by CVaR with different parameter under Black-Scholes model



We will use $\alpha = 0.95$ on the following experiment.

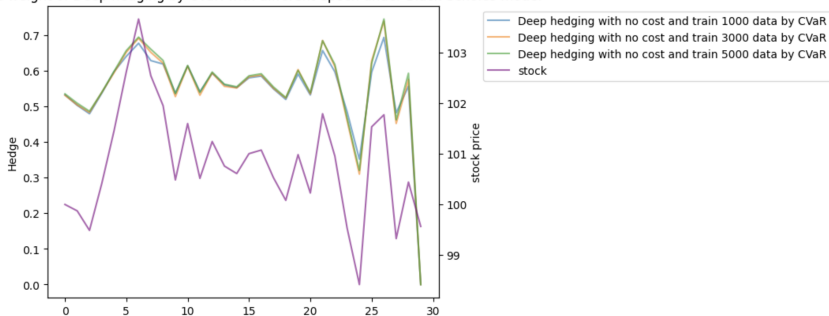
Deep hedging with different epoch

We know some function converge very slowly, so we need to insure they have enough training epoch to converge the best solution.

We compare with training epoch 1000, 3000, 5000, 10000

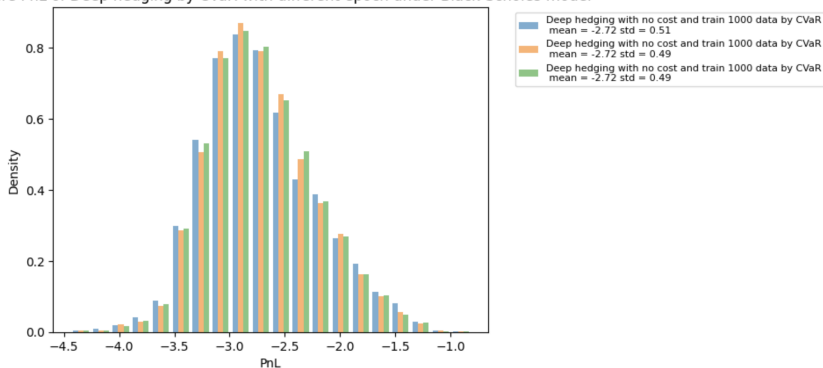
Deep hedging(CVaR) with different epoch

Compare weight of Deep hedging by CVaR with different epoch under Black-Scholes model



Deep hedging(CVaR) with different epoch

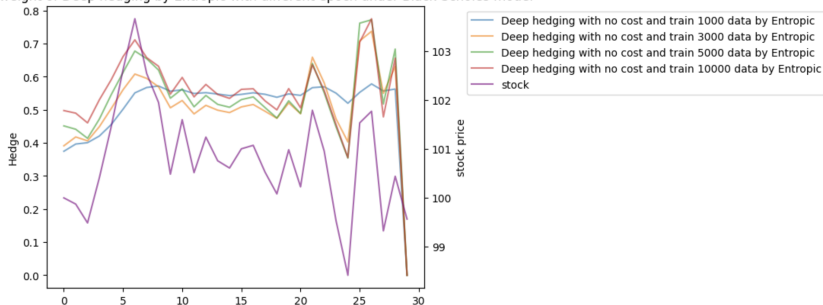
Compare PnL of Deep hedging by CVaR with different epoch under Black-Scholes model



We will use training epoch 1000 on the following experiment for CVaR.

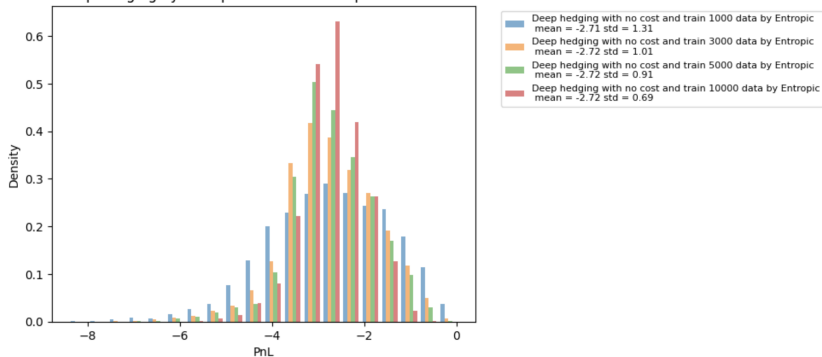
Deep hedging(Entropic) with different epoch

Compare weight of Deep hedging by Entropic with different epoch under Black-Scholes model



Deep hedging(Entropic) with different epoch

Compare PnL of Deep hedging by Entropic with different epoch under Black-Scholes model



Deep hedging(Entropic) with different epoch

Loss after training

- 1000 epochs: 265.6
- 3000 epochs: 237.1
- 5000 epochs: 204.2
- 10000 epochs: 20.4

We will use training epoch 10000 on the following experiment for Entropic.

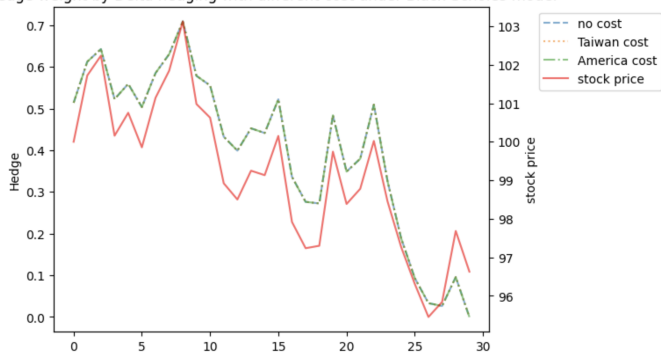
Different transaction costs

For BS-model

- Delta hedging with different cost
- Deep Hedging (CVaR) with Different Cost
- Deep Hedging (Entropic) with Different Cost
- No Cost with Different Strategies
- Taiwan Cost with Different Strategies
- American Cost with Different Strategies

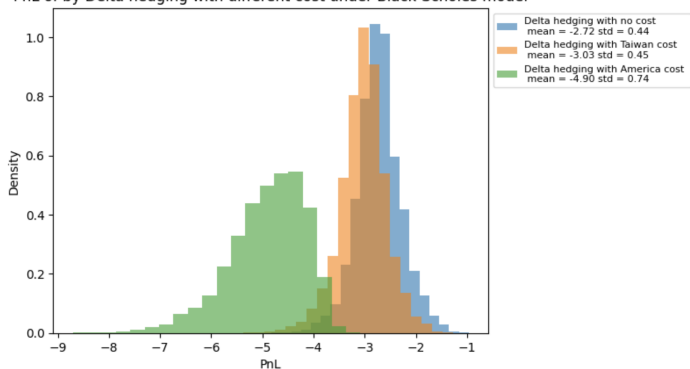
Delta hedging with different cost

Hedge weight by Delta hedging with different cost under Black-Scholes model



Delta hedging with different cost

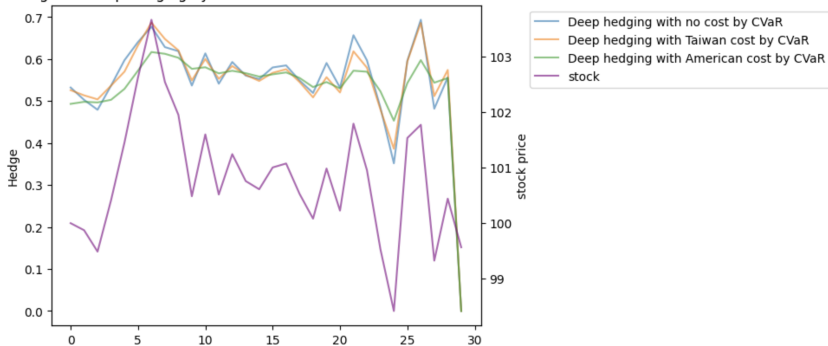
PnL of by Delta hedging with different cost under Black-Scholes model



Delta Hedging strategy shows a clear sensitivity to transaction costs.

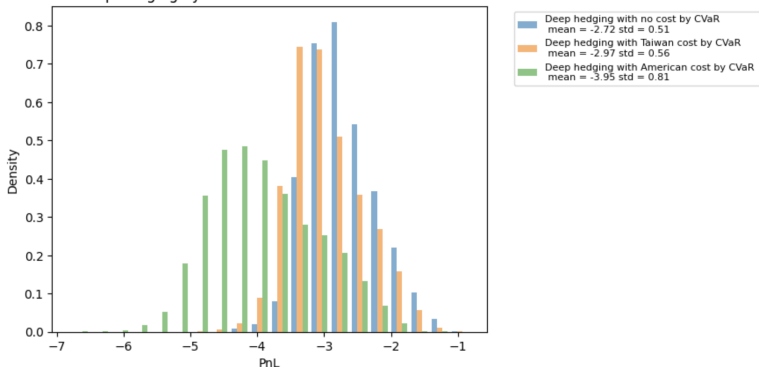
Deep Hedging (CVaR) with Different Cost

Compare weight of Deep hedging by CVaR with different cost under Black-Scholes model



Deep Hedging (CVaR) with Different Cost

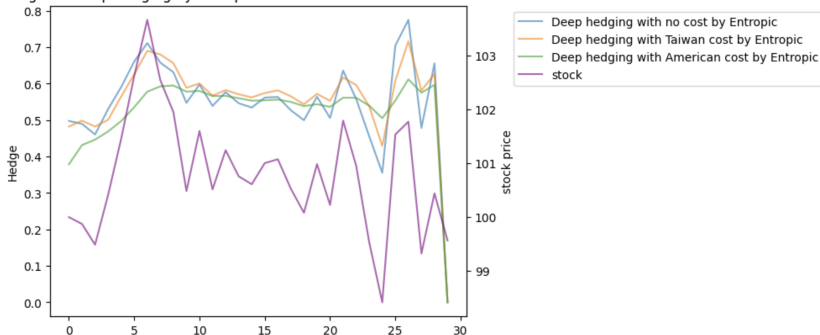
Compare PnL of Deep hedging by CVaR with different cost under Black-Scholes model



Deep Hedging (CVaR) strategy shows the higher the fee ratio, the lower the PnL, and the fluctuation in weights also becomes smaller.

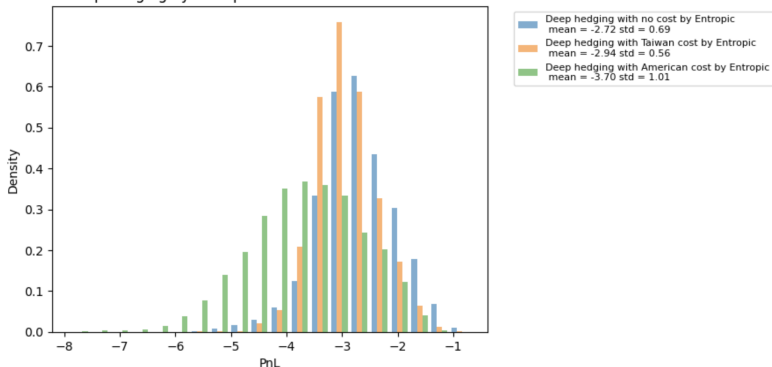
Deep Hedging (Entropic) with Different Cost

Compare weight of Deep hedging by Entropic with different cost under Black-Scholes model



Deep Hedging (Entropic) with Different Cost

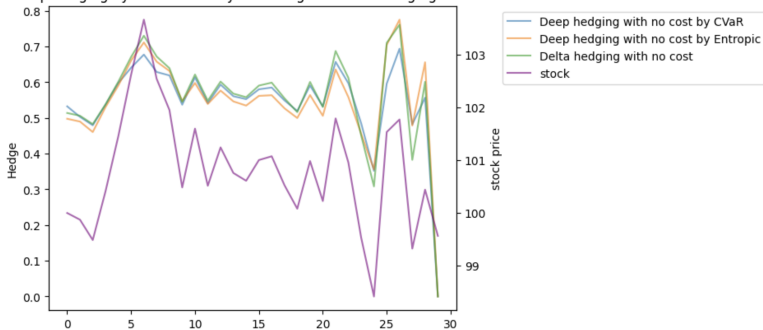
Compare PnL of Deep hedging by Entropic with different cost under Black-Scholes model



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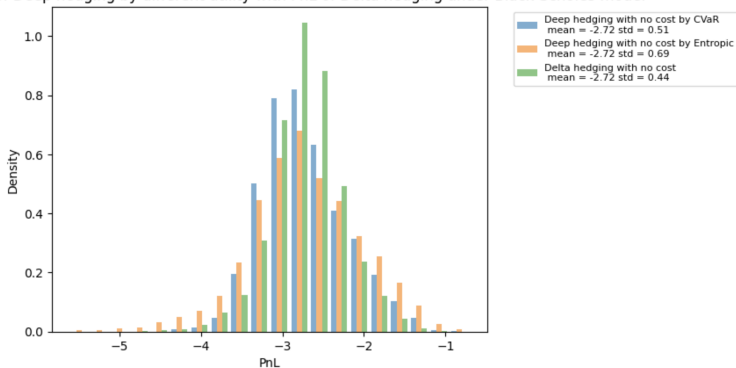
No Cost with Different Strategies

Compare weight of Deep hedging by different utility with weight of Delta hedging under Black-Scholes model



No Cost with Different Strategies

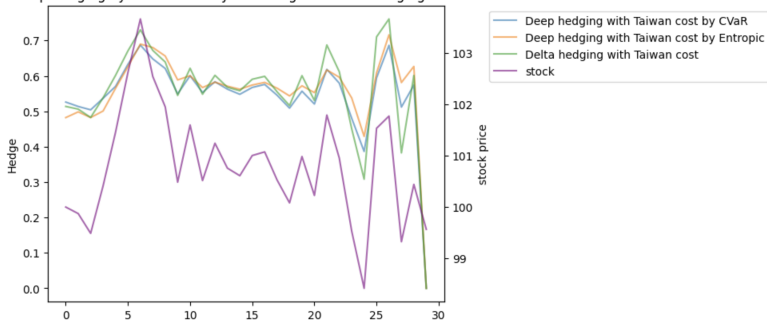
Compare PnL of Deep hedging by different utility with PnL of Delta hedging under Black-Scholes model



Delta Hedging strategy shows the better outcome.

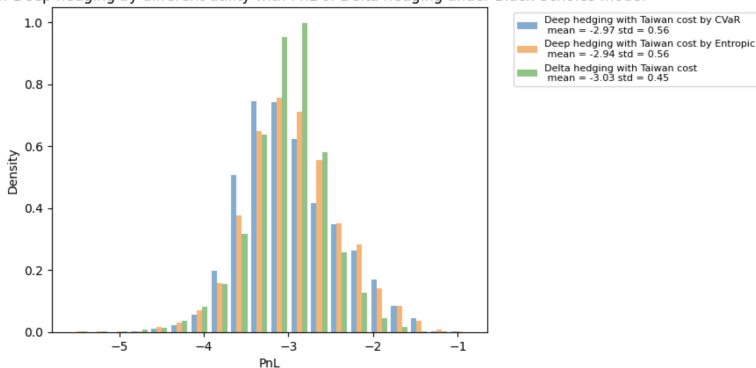
Taiwan Cost with Different Strategies

Compare weight of Deep hedging by different utility with weight of Delta hedging under Black-Scholes model



Taiwan Cost with Different Strategies

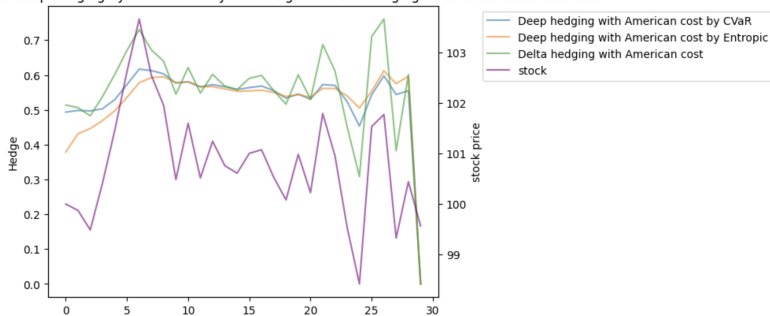
Compare PnL of Deep hedging by different utility with PnL of Delta hedging under Black-Scholes model



Deep Hedging strategy shows the better outcome.

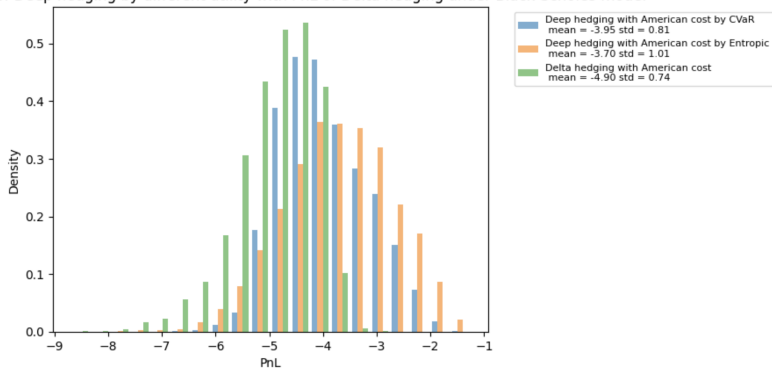
American Cost with Different Strategies

Compare weight of Deep hedging by different utility with weight of Delta hedging under Black-Scholes model



American Cost with Different Strategies

Compare PnL of Deep hedging by different utility with PnL of Delta hedging under Black-Scholes model



Deep Hedging strategy shows the better outcome.

Conclusion

- Under transaction costs, no matter which strategies we choose, the PnL decreases.
- With higher transaction costs, the changes in trading volume for deep hedging become less frequent.
- Under transaction costs, the performance of delta hedging becomes worse compared to the other two strategies.

Different transaction costs

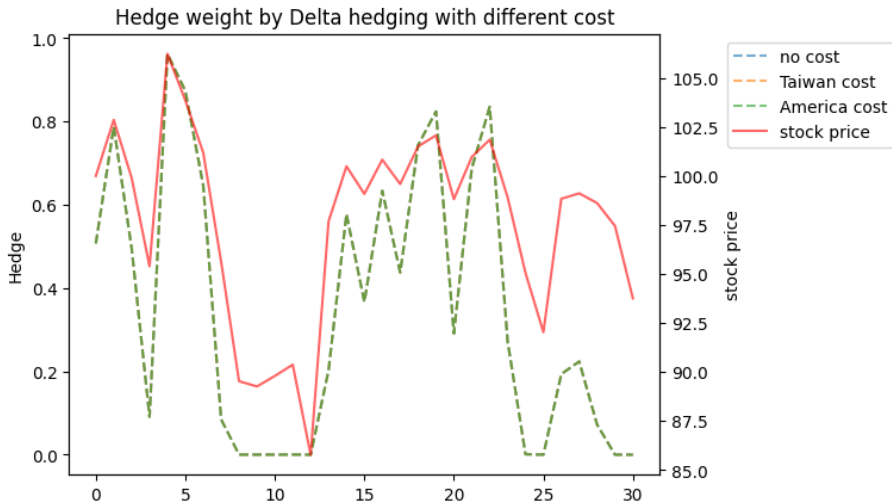
For AAPL History Data

- Delta hedging with different cost
- Deep Hedging (CVaR) with Different Cost
- Deep Hedging (Entropic) with Different Cost
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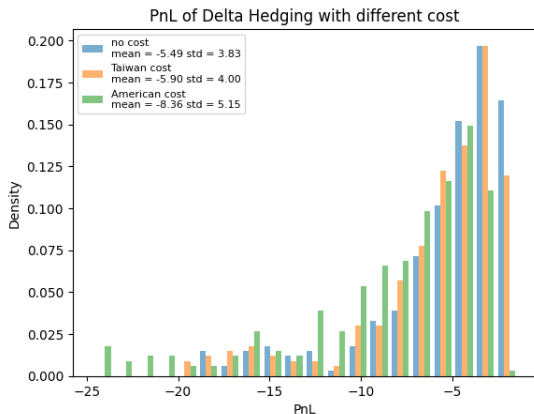
AAPL History Data



Delta hedging with different cost



Delta hedging with different cost

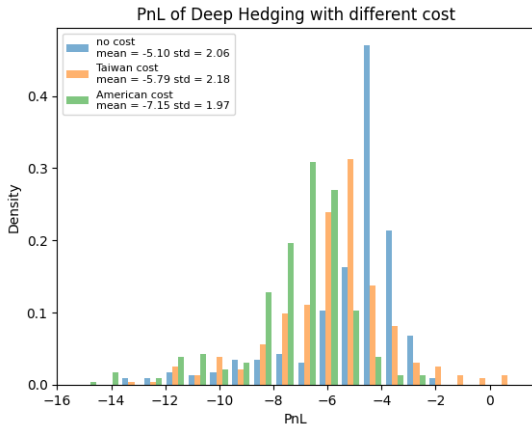


Delta Hedging strategy shows a clear sensitivity to transaction costs.

Deep Hedging (CVaR) with Different Cost

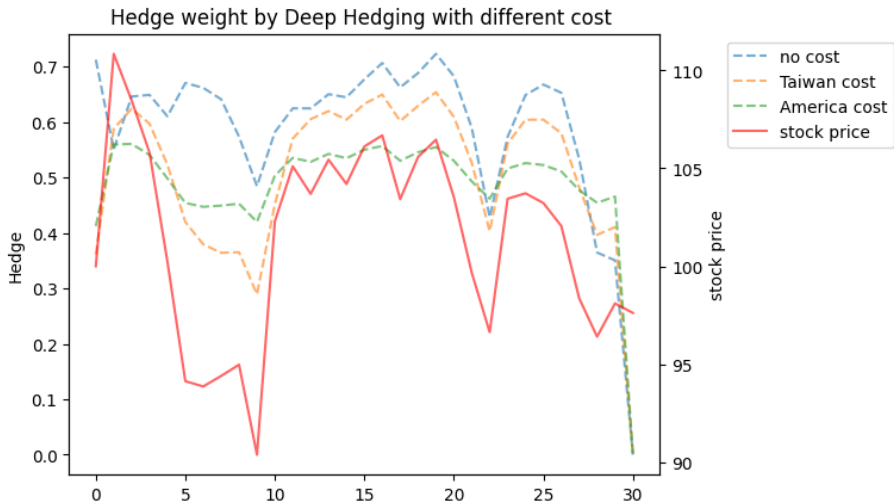


Deep Hedging (CVaR) with Different Cost

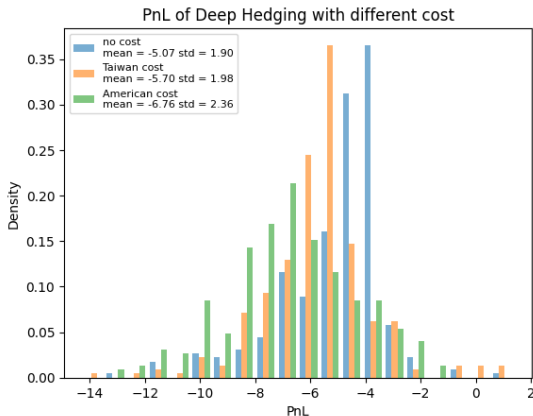


Deep Hedging (CVaR) strategy shows the higher the fee ratio, the lower the PnL, and the fluctuation in weights also becomes smaller.

Deep Hedging (Entropic) with Different Cost

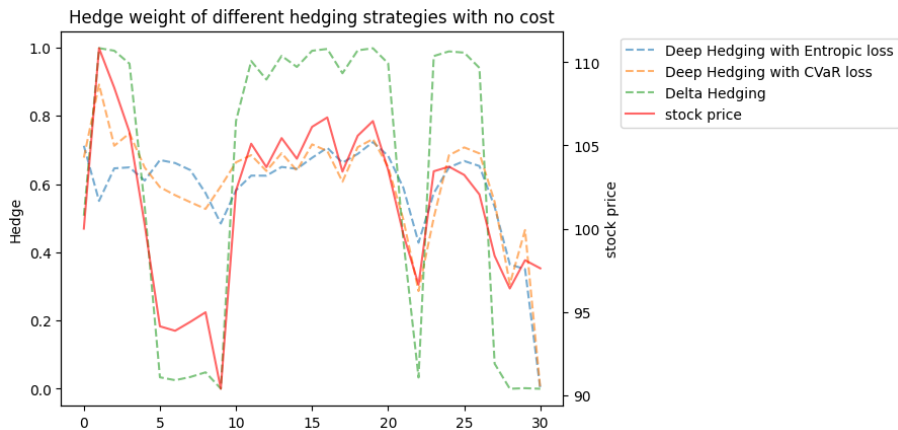


Deep Hedging (Entropic) with Different Cost

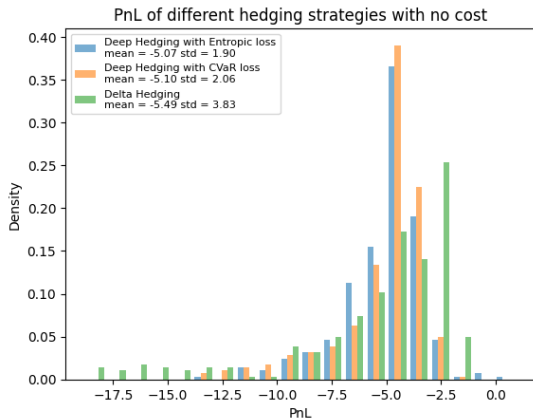


Deep Hedging (Entropic) strategy shows the higher the fee ratio, the lower the PnL, and the fluctuation in weights also becomes smaller.

No Cost with Different Strategies



No Cost with Different Strategies

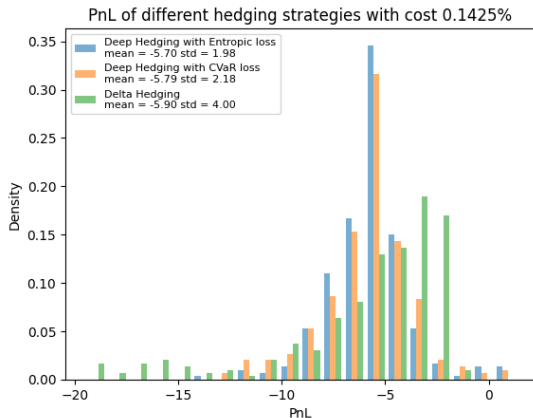


Deep Hedging strategy shows the better outcome.

Taiwan Cost with Different Strategies

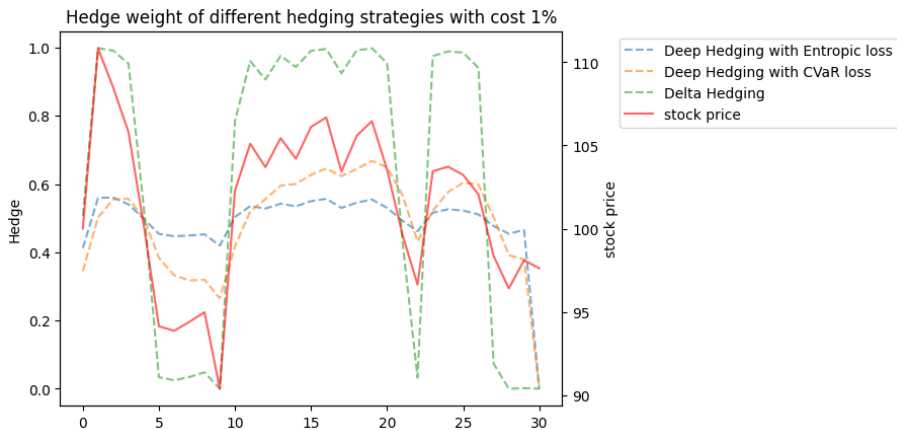


Taiwan Cost with Different Strategies

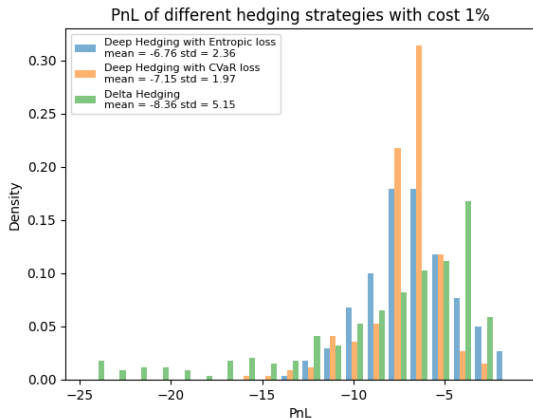


Deep Hedging strategy shows the better outcome.

American Cost with Different Strategies



American Cost with Different Strategies



Deep Hedging strategy shows the better outcome.

Conclusion

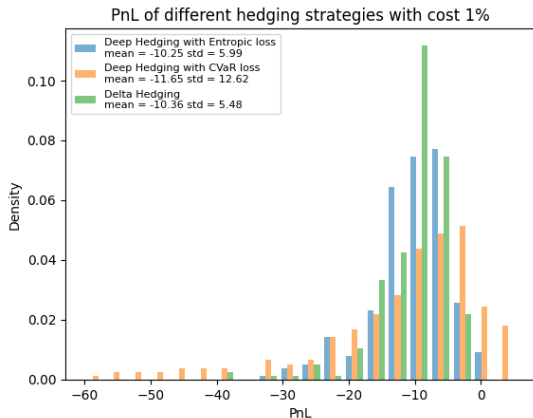
- Under transaction costs, no matter which strategies we choose, the PnL decreases.
- With higher transaction costs, the changes in trading volume for deep hedging become less frequent.
- Under transaction costs, the performance of delta hedging becomes worse compared to the other two strategies.

Deep Hedging strategies offer superior performance and resilience compared to traditional Delta Hedging, particularly in the presence of transaction costs.

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Observation: Deep hedging underperforms Black-Scholes delta hedging.



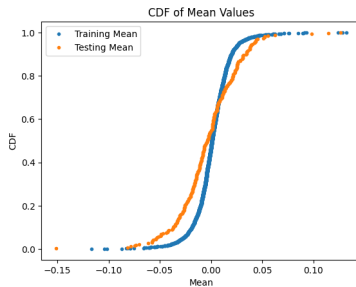
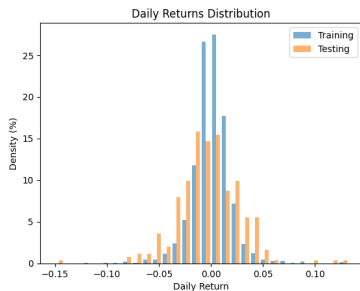
Amazon stock data

Action: Changed the time period of the Amazon data.



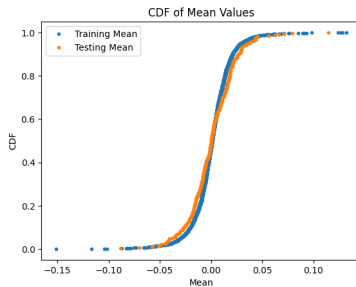
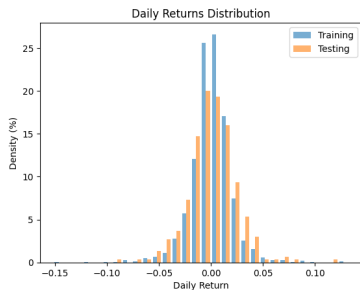
Amazon stock data

Observation: The distribution of daily returns differs significantly between the training and testing datasets.



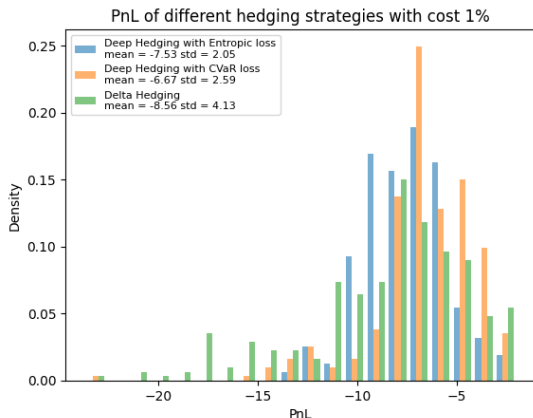
Amazon stock data

Action: Reevaluated the distribution differences between the training and testing datasets. The differences are not substantial.



Amazon stock data

Observation: In the revised period, deep hedging outperforms delta hedging, similar to the previous results with Apple stock data.



Conclusion

- Deep Hedging strategies may underperform traditional delta hedging in unforeseen market conditions.
- Deep Hedging strategies perform optimally when the training and testing data distributions are similar.
- Proper preprocessing of training data is essential, with potential use of GANs to generate synthetic data for better performance.